Which of the following statements are equivalent? Which ones are correctly indented?

```
if (i > 0) if
(j > 0)
x = 0; else
if (k > 0) y = 0;
else z = 0;
```

(a)

```
if (i > 0) {
  if (j > 0)
    x = 0;
  else if (k > 0)
    y = 0;
}
else
  z = 0;
```

(b)

```
if (i > 0)
  if (j > 0)
    x = 0;
  else if (k > 0)
    y = 0;
  else
    z = 0;
```

(c)

```
if (i > 0)
  if (j > 0)
    x = 0;
  else if (k > 0)
    y = 0;
else
  z = 0;
```

(d)

The `for` loop on the left is converted into the `while` loop on the right. What is wrong? Correct it.

```
for (int i = 0; i < 4; i++)
{
  if (i % 3 == 0) continue;
  sum += i;
}
```

Converted → Wrong conversion

```
int i = 0;
while (i < 4)
{
  if (i % 3 == 0) continue;
  sum += i;
  i++;
}
```

Identify and fix the syntax errors in (a) and the logic errors in (b).

```
for (int i = 0; i < 10; i++);
  sum += i;

if (i < j);
  cout << i
else
  cout << j;

while (j < 10);
{
  j++;
};

do
{
  j++;
} while (j < 10)
```

(a)

```
int total = 0, num = 0;

do
{
  // Read the next data
  cout << "Enter an int value, " <<
    "\nexit if the input is 0: ";
  int num;
  cin >> num;

  total += num;
} while (num != 0);

cout << "Total is " << total << endl;
```

(b)

What is the keyword **break** for? What is the keyword **continue** for? Will the following program terminate? If so, give the output.

```
int balance = 1000;
while (true)
{
  if (balance < 9)
    break;
  balance = balance - 9;
}

cout << "Balance is " <<
  balance << endl;
```

(a)

```
int balance = 1000;
while (true)
{
  if (balance < 9)
    continue;
  balance = balance - 9;
}

cout << "Balance is "
  << balance << endl;
```

(b)